Achromatic Polarization Rotation in Black Hole Images: A Falsifiable Event Horizon Telescope Test for New Physics

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Abstract

Polarimetric EHT observations of M87* and Sagittarius A* show dynamic and partly surprising structures. We propose a simple yet rigorous test for new physics: a frequency independent rotation of linear polarization produced by an axial scalar field a with a fixed axion photon coupling factor. The effect occurs locally in the emission region near the event horizon and is *achromatic*. Its normalization contains no free parameters, because the coupling $g_{a\gamma\gamma}$ is topologically fixed by $c_3 = 1/(8\pi)$ and the global normalization, already tested cosmologically, is set by the geometric invariant φ_0. We provide a complete test configuration for the EHT, a clean analysis pathway with null tests, and a discussion of the payoff: a confirmation would simultaneously (i) explain the observed polarization anomalies, (ii) support a unified field equation with torsion, and (iii) establish new measurable links between near field polarimetry and cosmic birefringence.[1]

1 Scientific background and motivation

1.1 Structural premises

The theory used here is based on two fixed invariants: a topological one $c_3 = 1/(8\pi)$ and a geometric one $\varphi_0 \approx 0.05317$. Together with the Standard Model content they lead (via a cubic fixed point condition) to a compact unified field equation with torsion as well as to a unique normalization of the axion photon coupling. The point is essential: neither $g_{a\gamma\gamma}$ nor the cosmic rotation need to be tuned; they are *given*.[2]

1.2 Cosmic anchoring of the normalization

From the modified Maxwell system one obtains a uniform rotation of linear polarization:

$$\beta = 2c_3\,\Delta a = \frac{1}{4\pi}\,\Delta a.$$

With the minimal choice $\Delta a = \varphi_0$, the closed parameter free prediction for cosmic birefringence follows

$$\beta_{\text{th}} = \frac{\varphi_0}{4\pi} \simeq 0.2427^\circ,$$

which is consistent with PR4 analyses of Planck (EB mixing) at the level of 0.5 to 1.7σ. Thus the *global* normalization is already anchored close to observations and cannot be freely varied in the near field test.[3]

[1] Unified Field Equation with torsion, axion modified Maxwell equations, and the resulting rotation formula $d\beta/d\eta = 2c_3\,da/d\eta$ are given in the UFE shorthand. Achromaticity and the identification $g_{a\gamma\gamma} = -4c_3$ follow directly. [oai$_c$$itation : 0Unified_Field_Equation.pdf$](sediment : //file$_0$000000079e0620aaa4bb73739da2279)

[2] Topological and geometric inputs, cubic fixed point logic, and the resulting axion modified Maxwell sector are summarized in the UFE note; there $g_{a\gamma\gamma} = -4c_3$ is also derived. [oai$_c$$itation : 1Unified_Field_Equation.pdf$](sediment : //file$_0$000000079e0620aaa4bb73739da2279)

[3] Derivation of the relation $\beta = \varphi_0/(4\pi)$ and a tabulated comparison with PR4 values are documented in the UFE note. [oai$_c$$itation : 2Unified_Field_Equation.pdf$](sediment : //file$_0$000000079e0620aaa4bb73739da2279)

1.3 Geometric self consistency

The same electromagnetic energy that drives the logarithmic scale in the fixed point couples back minimally into geometry on the orientable double cover. This enforces a linear response $\varphi_0(\alpha) = \varphi_{\text{tree}} + \delta_{\text{top}}(1 - 2\alpha)$ with no new number. The result is a sub ppm closure of the cubic equation for α with no impact on the birefringence prediction.[4] This closure is experimentally relevant because it supports achromaticity as a fixed signature type, independent of plasma details.

2 Locally near the black hole: prediction

2.1 Achromatic rotation through $a(r)$

In the environment of a rotating hole in an external magnetic field an effective electric charge $Q_e^{\text{eff}} \simeq 2a_* B_0 M^2$ arises through the known Wald mechanism; the magnetosphere provides an effective magnetic flux quantity Q_m^{eff}.[5] For a photon packet emitted at radius r_{em} one obtains along the geodesic

$$\beta_{\text{BH}}(r_{\text{em}} \to \infty) = \frac{c_3}{32\pi^3} \frac{Q_e^{\text{eff}} Q_m^{\text{eff}}}{r_{\text{em}}^2} = \frac{1}{256\pi^4} \frac{Q_e^{\text{eff}} Q_m^{\text{eff}}}{r_{\text{em}}^2}, \tag{1}$$

where the second form uses $c_3 = 1/(8\pi)$. The key features are:

- *Achromaticity*: no λ^2 term, in contrast to Faraday rotation.
- *Radius dependence*: $1/r_{\text{em}}^2$ separates the near field effect from large scale depolarization.
- *Fixed normalization*: no free couplings; $g_{a\gamma\gamma}$ is fixed topologically.

2.2 Link to global tests

Equation (1) is not isolated but normalized by the same structure that yields the cosmic rotation $\beta_{\text{th}} = \varphi_0/(4\pi)$. The local test is therefore directly consistency linked to an already ongoing cosmological program; a positive measurement at M87* or Sgr A* would connect both levels.[6]

3 EHT test: configuration and analysis plan

3.1 Targets and frequencies

We propose observations of M87* and Sagittarius A* at 230 GHz and 345 GHz, preferably simultaneous or quasi simultaneous. These channels are already in EHT operations. The choice is methodologically central, because Faraday rotation scales as $\propto \lambda^2$, while the relevant contribution here is constant.

[4] Geometric self consistency on the orientable double cover and the numerical effect (-0.064 ppm) are shown in the short note. [oai$_c$*itation* : 3*ImprovedCubic.pdf*](*sediment* : *//file*$_0$0000000458*c*624390*f*707*b*5*e*66*b*099*c*)

[5] The dyonic computation class yields a closed solution with $a'(r) \neq 0$ once $E \cdot B \neq 0$. For real holes one formally replaces the monopole by Q_e^{eff} from Wald and interprets Q_m^{eff} as a magnetospheric flux quantity. The resulting rotation is achromatic and scales with $1/r_{\text{em}}^2$. Details, including the closed metric correction D/r^4, are given in UFE BH1. [oai$_c$*itation* : 4*BlackHoles.pdf*](*sediment* : *//file*$_0$00000008*f*98620*ab*77*c*0*ac*13910023*a*)

[6] The link from local achromaticity to cosmic EB mixing is documented throughout TFPT v1.0.8 and the UFE note, including the identification $g_{a\gamma\gamma} = -4c_3$ and the value β_{th}. [oai$_c$*itation* : 5*EliminatingK.pdf*](*sediment* : *//file*$_0$0000000*c*614620*a*90*f*0*ef*0685*a*4179*c*)[*oai*$_c$*itation* : 6*Unified*$_F$*ield*$_E$*quation.pdf*](*sediment* : *//file*$_0$000000079*e*0620*aaa*4*bb*73739*da*2279)

3.2 Image based analysis: intercept maps

The linear polarization angle χ can be fitted pixel by pixel against λ^2:

$$\chi(\boldsymbol{x}, \lambda^2) = \chi_0(\boldsymbol{x}) + \mathrm{RM}(\boldsymbol{x})\,\lambda^2 + \varepsilon. \tag{2}$$

Standard plasma says $\chi_0(\boldsymbol{x}) \equiv 0$. The axion driven near field effect requires instead a *non zero* intercept $\chi_0(\boldsymbol{x}) = \beta_{\mathrm{BH}}(\boldsymbol{x})$ whose spatial variation is characteristically modulated by $r_{\mathrm{em}}(\boldsymbol{x})$. The EHT specific procedure:

1. Polarimetric calibration as usual; absolute EVPA anchors through the ALMA sub array or established standards (no methodological novelty).
2. Reconstruction of Stokes Q, U at 230 GHz and 345 GHz; generation of EVPA maps.
3. Joint fit of $\chi(\boldsymbol{x}, \lambda^2)$ across both bands for all image points $\boldsymbol{x}$ above an SNR threshold; the result is intercept maps $\chi_0(\boldsymbol{x})$ and RM maps.
4. Diagnostic maps: (i) achromaticity test $\Delta\chi(\boldsymbol{x})$ between 230 GHz and 345 GHz; (ii) correlation of $\chi_0(\boldsymbol{x})$ with a geometric radius indicator $r_{\mathrm{em}}(\boldsymbol{x})$ from the image reconstruction (for example ring radius, photon sphere proxies).

A *significantly* non zero intercept over connected image regions and across epochs, plus the $1/r^2$ tendency along radial gradients, is the sought signature.

3.3 Array based robustness: closure null tests

To separate from instrumental offsets and calibration degeneracies we recommend:

- **Achromaticity**: A fit with $\chi_0 \neq 0$ and free RM must describe the data visibly better than a fit that enforces $\chi_0 \equiv 0$. The gain should be coherent across bands.
- **Sign flip coherence**: If the effective orientation of E and B in the magnetosphere inverts between epochs, the sign of χ_0 must coherently invert as well. This test counters constant instrumental offsets.[7]
- **Scale decoupling**: Compare the intercept map with the RM map. Strong achromaticity together with moderate RM values argues against pure plasma processes.

3.4 Time budget and feasibility

No special scheduling beyond the existing EHT program is needed. The analysis uses standard products (Stokes images, EVPA maps) and requires only an additional per pixel intercept estimation plus simple null tests. The attraction lies in the clarity of the test: *either* a non vanishing coherent intercept with achromaticity and a $1/r^2$ trend *or* a null result that can be translated strictly into model parameters.

4 Expected results and interpretation

4.1 Primary finding

A statistically significant non vanishing intercept $\chi_0(\boldsymbol{x})$ together with good λ^2 linearity of the RM terms would be the primary discovery. The spatial pattern of $\chi_0(\boldsymbol{x})$ along radial gradients should reflect a $1/r_{\mathrm{em}}^2$ trend.

[7] The sign flip prediction as a consequence of the effective $E \cdot B$ orientation is formulated as H2 in the BH1 note. [oai$_c itation : 7BlackHoles.pdf](sediment : //file_0 00000008f98620ab77c0ac13910023a)$

4.2 Quantitative control

The size of χ_0 can be compared via equation (1) with *effective* quantities. Q_e^{eff} follows from Wald $[2a_* B_0 M^2]$, Q_m^{eff} can be estimated as a magnetospheric flux quantity from MHD models or poloidal field inference. The normalization is fixed by c_3, a fit is not required.[8]

4.3 Negative result as a hard bound

A consistent null finding (intercept with $|\chi_0| \ll$ error bars, no sign flip, no achromaticity) across multiple epochs and both frequencies sets a direct upper limit on the product $Q_e^{\text{eff}} Q_m^{\text{eff}}$ and thus on the strength of the near field effect driven by $E \cdot B$ within a theory whose coupling $g_{a\gamma\gamma}$ is *not* free. The theory thus remains sharply falsifiable.

5 Why this test is worthwhile

5.1 Physics gain upon confirmation

A positive measurement would:

1. support the existence of an axionic background in the near field of black holes with fixed normalization,
2. anchor the unified field equation with torsion in the electromagnetic sector through direct observation,
3. build a bridge to the cosmic EB signature that is already seen consistent with the prediction β_{th}.[9]

5.2 Robustness with respect to standard explanations

Plasma processes produce rotation measure dependent, frequency selective effects ($\propto \lambda^2$). The finding required here is in essence *achromatic* and is furthermore characterized by sign flip coherence. Both are difficult to mimic through instrumental or plasma systematics.

5.3 Coherence with the metric limit

In the torsion vacuum the underlying equation reduces exactly to the Einstein Hilbert normalization ($\kappa^2 = 8\pi G$). The theory is therefore consistent in the relevant limit and violates no classic tests of general relativity; the proposed test provides genuine additional information rather than a contradiction.[10]

6 Practical notes for the collaboration

6.1 Data products

In addition to the usual Stokes reconstructions we propose three further lightweight products:

[8] The interplay of local prediction, Wald replacement, and global normalization is given explicitly in the BH1 note; there the small but fixed metric correction D/r^4 is also documented, which can serve as a consistency check. $[\text{oai}_c itation : 8BlackHoles.pdf](sediment : //file_0 00000008f98620ab77c0ac13910023a)$

[9] A coherent presentation of the full chain from c_3, φ_0 through the cubic fixed point logic to near field and far field signatures is given in TFPT v1.0.8. $[\text{oai}_c itation : 9EliminatingK.pdf](sediment : //file_0 0000000c614620a90f0ef0685a4179c)$

[10] The fixing of the gravitational coupling κ^2 from (φ_0, c_3) and the explicit recovery of the Einstein limit are shown in Note H1. $[\text{oai}_c itation : 10FiveProblems.pdf](sediment : //file_0 000000076806246a7950db988458daa)$

- Intercept map $\chi_0(\boldsymbol{x})$ from a joint fit across 230 GHz and 345 GHz.
- Achromaticity diagnostic $\Delta\chi(\boldsymbol{x})$ and RM map for direct comparison.
- Epoch by epoch comparison of the sign structure of $\chi_0(\boldsymbol{x})$ (search for sign flips).

6.2 Control sources and systematics

EVPA calibrations can be performed as usual with known polarization standards. For achromaticity the relative consistency between 230 GHz and 345 GHz is sufficient; absolute levels serve as plausibility anchors.

7 Conclusion

The proposed test does not require exotic observations but only an additional well defined analysis of already planned EHT data. If positive, it opens a direct path to new physics with fixed coupling and no free parameters. If negative, it yields hard bounds that can be translated into model parameters. In both cases the information gain is high and the effort is moderate.

Pointers to theory sources within this manuscript
The core formulas and normalizations are given in the following documents: UFE shorthand with $g_{a\gamma\gamma} = -4c_3$ and $\beta = \varphi_0/(4\pi)$ (including PR4 comparison), geometric self consistency on the double cover, TFPT v1.0.8 as an overall overview, BH1 note with the closed near field solution, and Note H1 on the Einstein limit.[11]

[11] See the footnotes in the text: UFE and cosmic birefringence [oai$_c$$itation$: $11Unified_Field_Equation.pdf](sediment : //file_0000000079e0620aaa4bb73739da2279); geometricselfconsistency[oai_citation : 12ImprovedCubic.pdf](sediment : //file_0000000458c624390f707b5e66b099c); TFPTv1.0.8[oai_citation : 13EliminatingK.pdf](sediment : //file_0000000c614620a90f0ef0685a4179c); BH1nearfieldsolution[oai_citation : 14BlackHoles.pdf](sediment : //file_00000008f98620ab77c0ac13910023a); Einsteinlimit[oai_citation : 15FiveProblems.pdf](sediment : //file_000000076806246a7950db988458daa)$.